Topic 5: Arms Race

Zeyn Jaswal, Azzaam Khan, Timothy Palin, Meredith Reeves, and Tony Xu
McMaster University

March 25, 2026

Overview

Scenario: Purple (P) and Green (G) are two countries competing in an arms race. Each is:

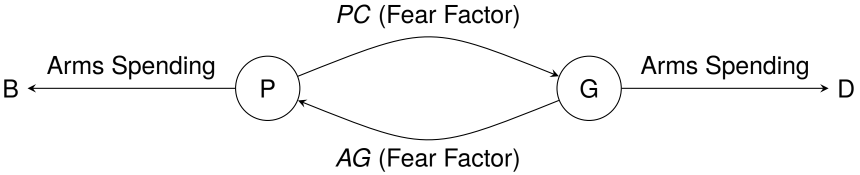
- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises

Research Question: What happens to the long-term spending of each country?

Base Model

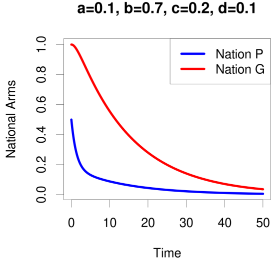
Model Equations

$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$



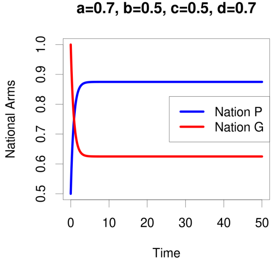
Long Term Behavior (Base Model)

$bd > ac$



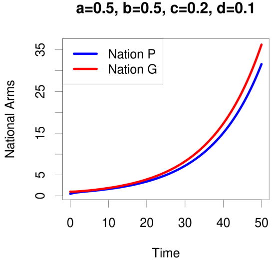
Complete mutual disarmament $(p^{\infty}, g^{\infty}) = (0, 0)$

$bd = ac$



Stabilized spending $(p^{\infty}, g^{\infty}) = \left(\frac{cp_0+bg_0}{a+c}, \frac{dp_0+ag_0}{a+c}\right)$

$bd < ac$

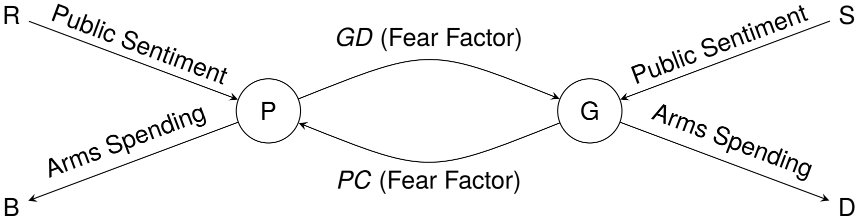


Perpetual arms race $(p^{\infty}, g^{\infty}) = (\infty, \infty)$

Grievances and Goodwill

Suppose P and G have underlying goodwill/grievances that influence their spending

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$



Long Term Behavior (Grievances and Goodwill)

- provide plots
- provide stability conditions and convergences
- This model will take on negative values so discuss that by either capping the value or other methods

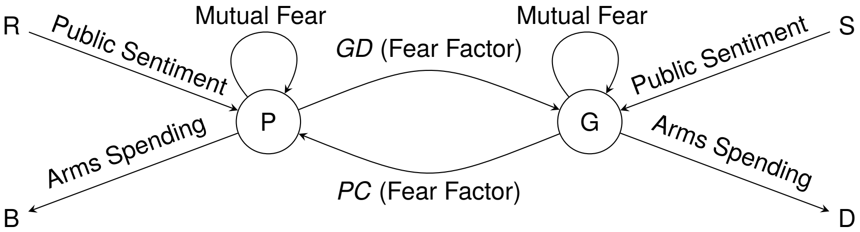
Mutual Logistic Fear

Limitation of Current Model: Does not account for the limited resources each country has

Solution: Introduce logistic growth

$$\frac{\partial p}{\partial t} = a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s$$



Long Term Behavior (Mutual Logistic Fear)

- provide plots
- provide stability conditions
- provide convergences (if possible)

Thank you for listening!

[0] Presentation theme by Ricardo B. Lourenco